

# ECO491/FIN591 – Problem Set 6: Option Full Valuation and Backtesting

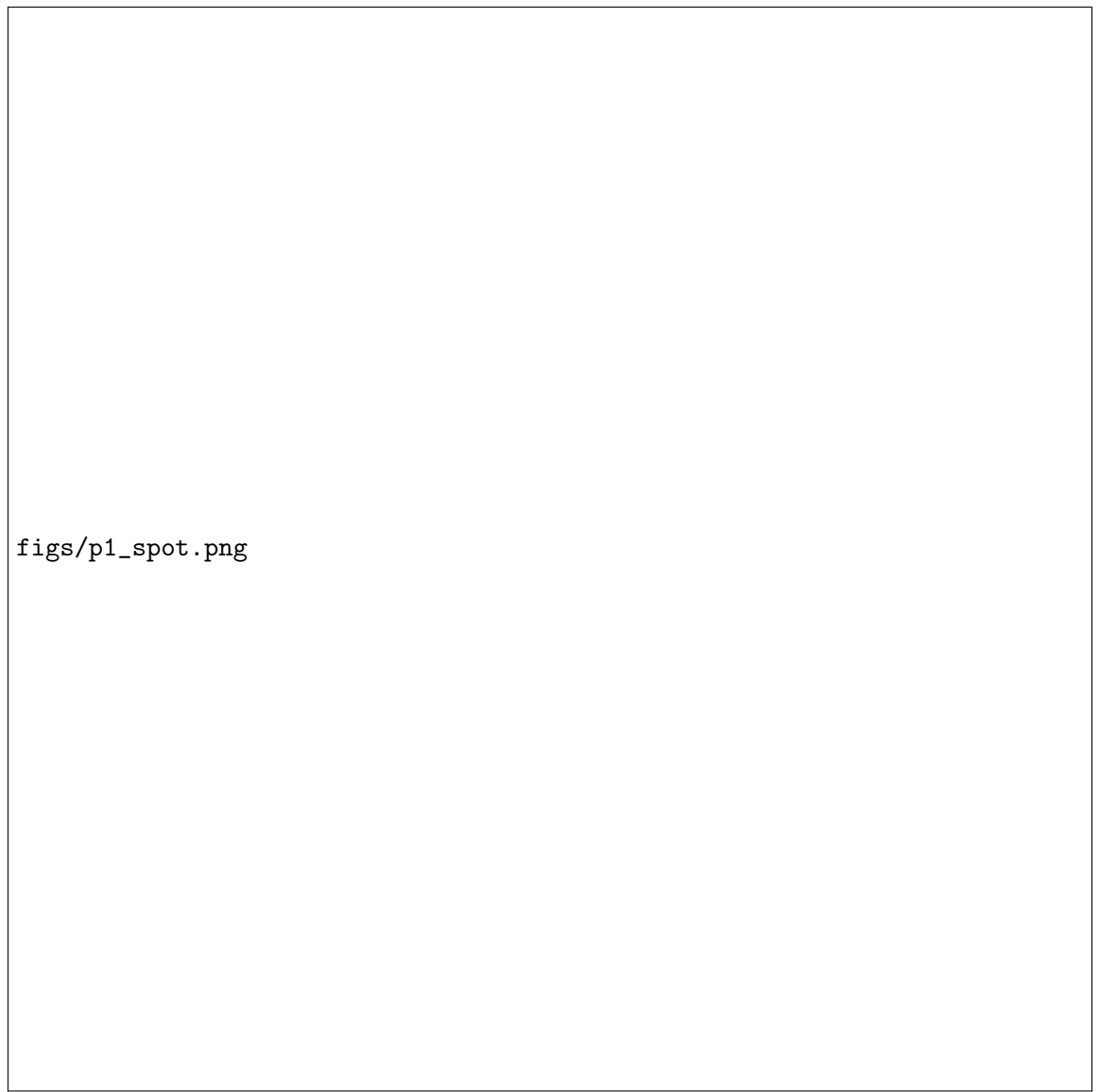
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## 1 1 Currency Option Risk

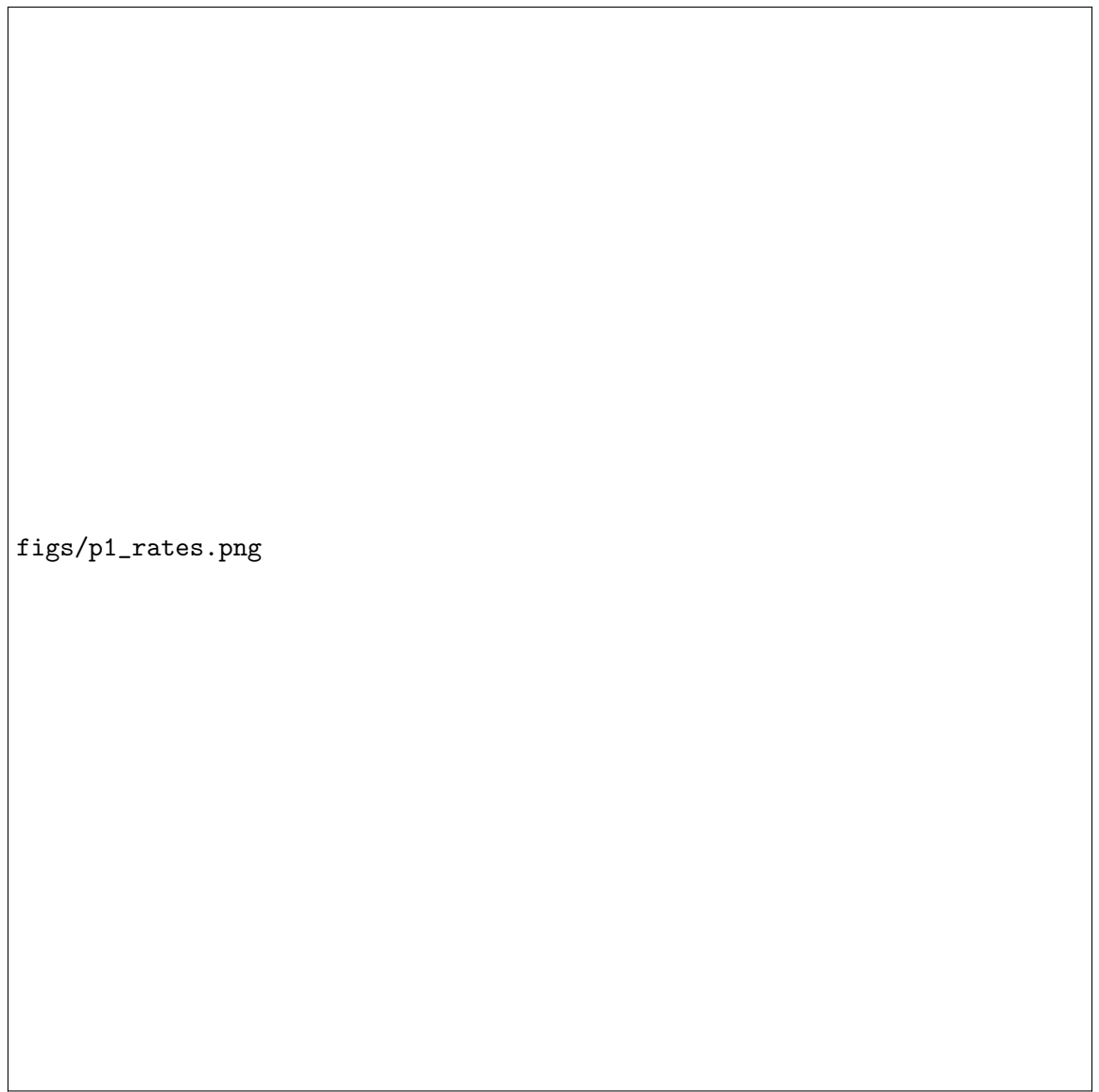
### 1.1 1.1 Risk factor plots and HS description

Risk factors are  $(S, r_d, r_f, \sigma)$ . Traditional HS creates daily shocks  $(\Delta \ln S, \Delta r_d, \Delta r_f, \Delta \sigma)$  from the historical series, applies each shock to today's factors, re-prices the option for each scenario (full valuation), and takes  $\text{VaR}_p = -Q_p(\Delta c)$ .



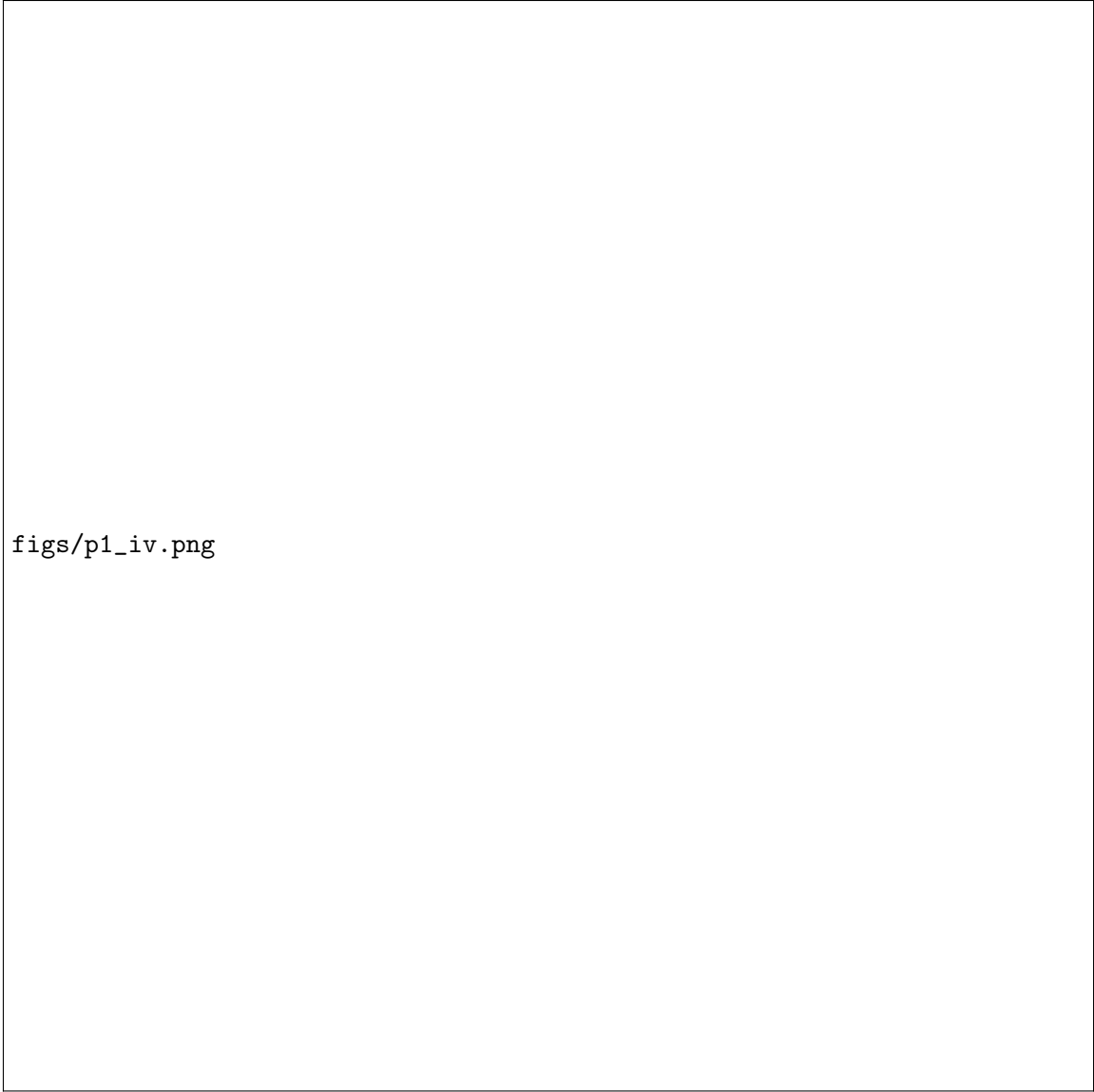
`figs/p1_spot.png`

Figure 1: USDBRL spot.



figs/p1\_rates.png

Figure 2: Interest rates.



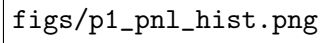
figs/p1\_iv.png

Figure 3: Implied volatility (3M).

### 1.2 1.2 HS VaR<sub>1%</sub> (full window, full valuation)

For  $S_0 = K = 3.919$  and  $T = 0.25$ , using the last-date  $(r_{d,0}, r_{f,0}, \sigma_0)$  from the spreadsheet:

$$\text{VaR}_{1\%} \approx 0.056437, \quad c_0 \approx 0.214807.$$

The figure is a histogram representing the distribution of P&L for a full valuation using the HS model. The x-axis represents the P&L values, and the y-axis represents the frequency or density of these values. The histogram shows a distribution of values, likely centered around a mean, with some spread indicating the range of possible outcomes. The file name 'figs/p1\_pnl\_hist.png' is visible in the top left corner of the plot area.

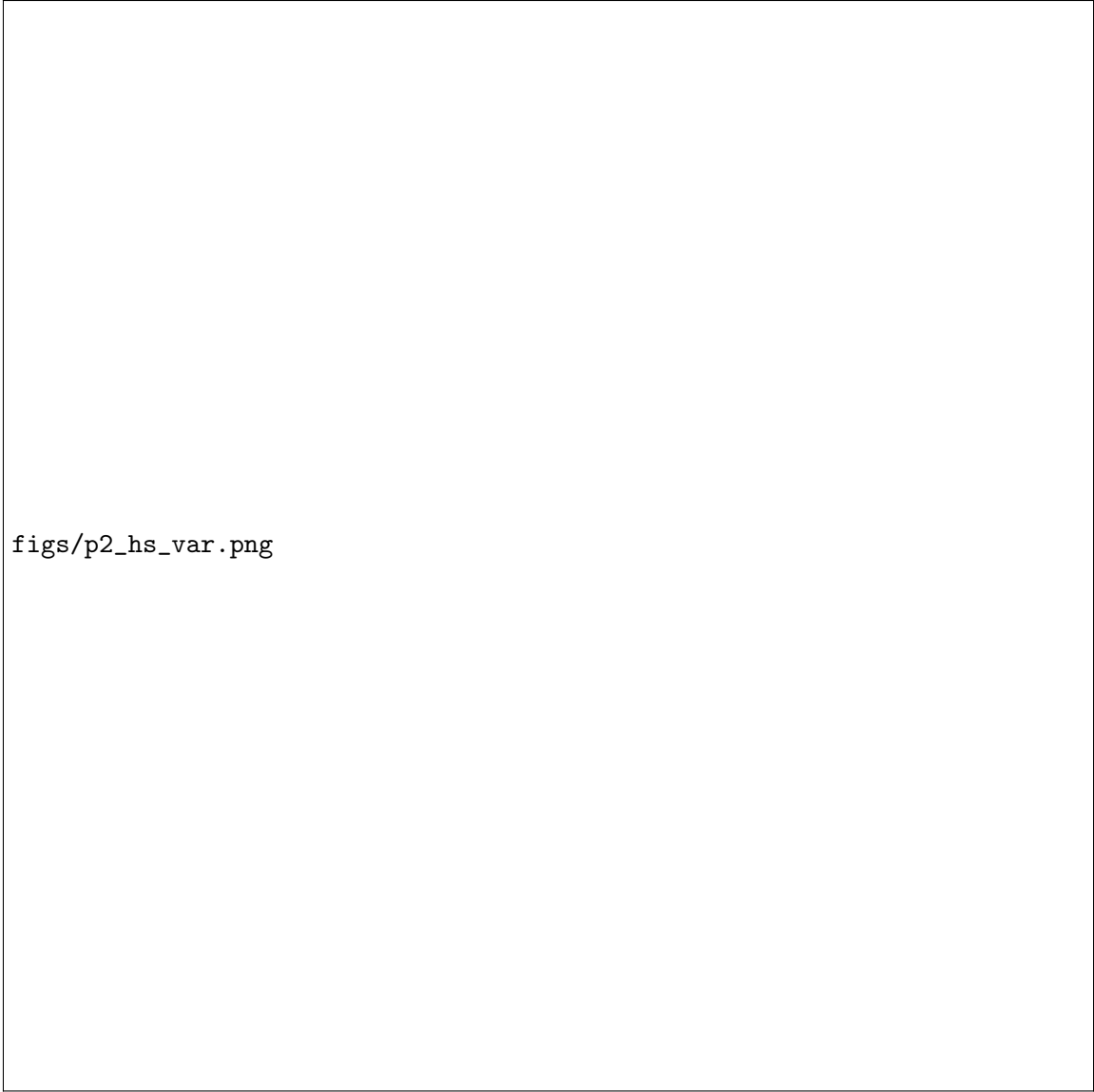
figs/p1\_pnl\_hist.png

Figure 4: HS P&L histogram (full valuation).

## 2 2 Backtesting Value at Risk (Apple)

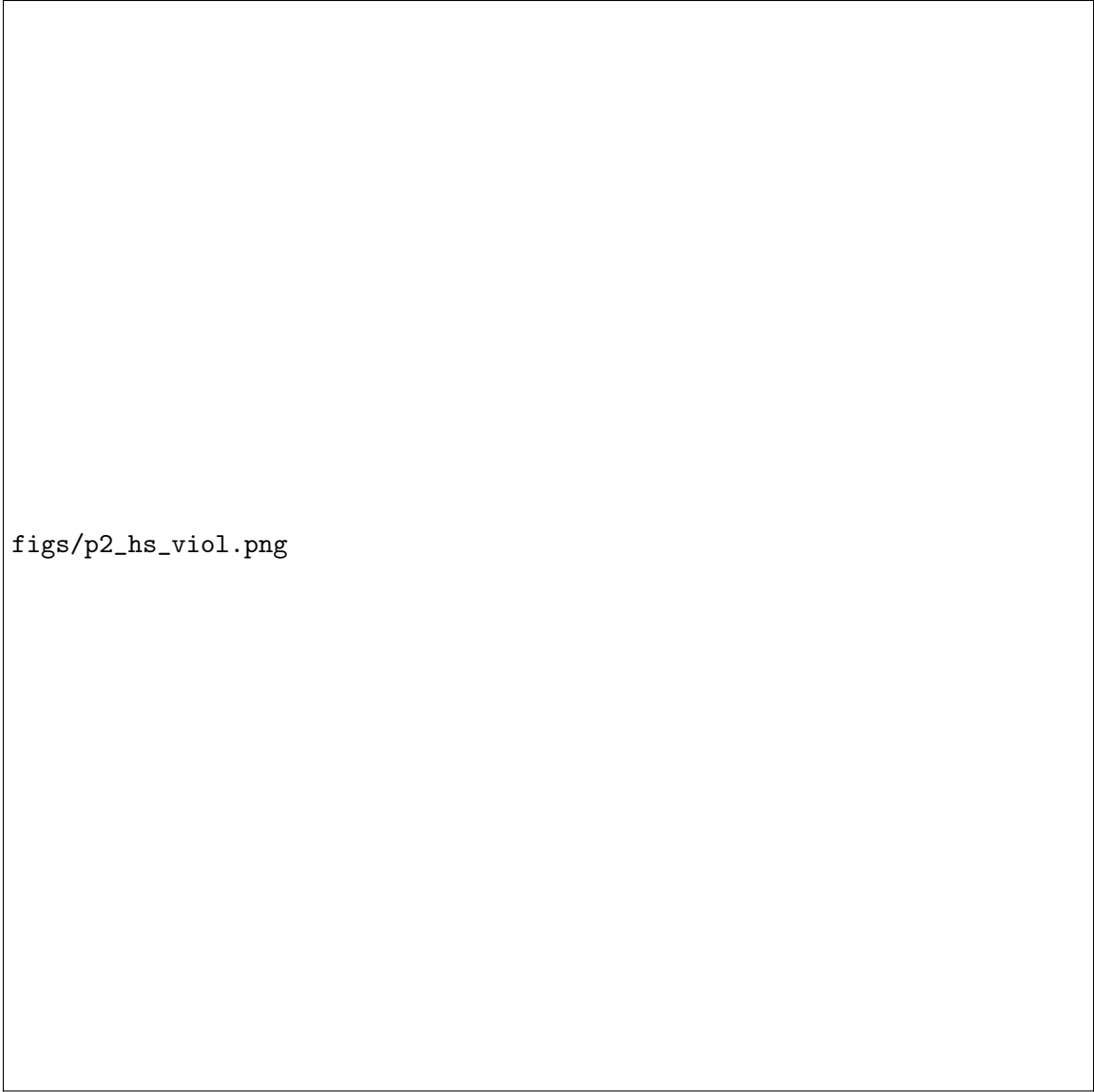
### 2.1 2.1 VaR series and hit sequences

Define hits by  $I_{t+1} = \mathbf{1}\{R_{t+1} < -\text{VaR}_{t+1}\}$ . Figures below show VaR and violations for HS (250d) and RiskMetrics EWMA ( $\lambda = 0.94$ ).



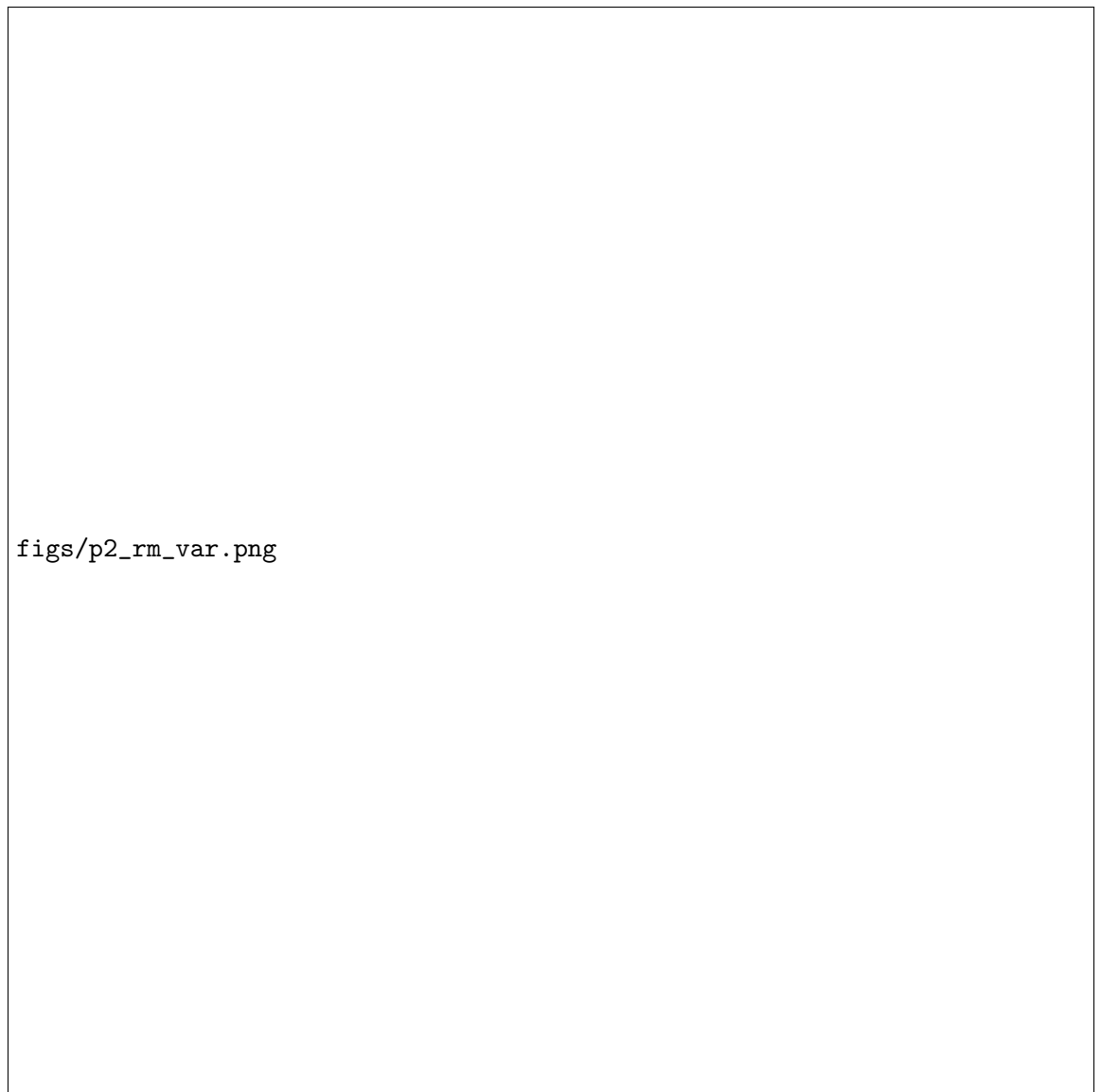
figs/p2\_hs\_var.png

Figure 5: HS VaR and returns (violations marked).



figs/p2\_hs\_viol.png

Figure 6: HS violations only.



figs/p2\_rm\_var.png

Figure 7: RiskMetrics VaR and returns (violations marked).



figs/p2\_rm\_viol.png

Figure 8: RiskMetrics violations only.

## 2.2 2.2 UC test

Critical value at 10%: 2.7055. Results (computed in the accompanying code):

| Method             | $T$  | $T_1$ | $\hat{\pi}$ | $LR_{uc}$ | Reject at 10%? |
|--------------------|------|-------|-------------|-----------|----------------|
| HS (250d)          | 1259 | 18    | 0.0143      | 2.0724    | No             |
| RiskMetrics (EWMA) | 1259 | 28    | 0.0222      | 14.1322   | Yes            |